Chapter 7 • Coordinate Geometry

Solved Exercise 7.3

1. Houses at (2,6) and (9,12), grid units in kilometres. Find distance.
$d = \sqrt{(7^2+6^2)}$
$= \sqrt{85} \approx 9.22$ km.

2. A straight trail runs from (5,7) to (15,3). Find its midpoint.
$M = (\frac{5+15}{2}, \frac{7+3}{2})$
$= (10,5)$.

3. Buildings at (10,8) and (4,3), coordinates in metres. Find distance.
$d = \sqrt{(6^2+5^2)}$
$= \sqrt{61} \approx 7.81$ m.

4. Delivery points (7,2) and (12,10), units in kilometres. Find distance.
$d = \sqrt{(5^2+8^2)}$
$= \sqrt{89} \approx 9.43$ km.

5. Find the midpoint of a race track from (3,9) to (9,13).
M=(6,11).

6. Find the midpoint of road endpoints A(3,4), B(7,10).
M=(5,7).

7. Use the textbook coordinate model for A(12°N,65°W) to B(20°N,45°W).
Take coordinate differences Δlat=8°, Δlong=20°. Planar distance=$\sqrt{(8^2+20^2)}$.
$4\sqrt{29} \approx 21.54$ coordinate-degree units. This is not a kilometre-scale great-circle distance.

8. Fence a rectangle with corners (0,0),(0,5),(8,5),(8,0).
Side lengths 8 and 5, so perimeter=2(8+5)=26 units.

9. Use the coordinate model from X(40°N,100°W) to Y(50°N,80°W).
Differences 10° and 20° give $\sqrt{(10^2+20^2)} = 10\sqrt{5} \approx 22.36$ coordinate-degree units.

10. Find the perimeter of the rectangle (3,1),(3,6),(8,6),(8,1).
Length 5 and width 5: perimeter=20 units.

11. Find fencing for rectangle A(0,0),B(5,0),C(5,3),D(0,3).
Perimeter $= 2(5+3)$
$= 16$ units.

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